

Literature status note for arXiv:math/0201187, Problem 1

Run `fa_banach_001`, lane 12

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Status. `literature_already_answered`. This packet records an exact later literature answer, not a new proof produced in the run.

Source question. In “Contractive projections and operator spaces”, arXiv:math/0201187, Neal and Russo ask as Problem 1 on PDF page 39:

What is $d_{cb}(H_n^k, R_n)$?

The same paper identifies H_n^n with the row Hilbert space R_n .

Answering source. In “Representation of contractively complemented Hilbertian operator spaces and the Fock space”, arXiv:math/0410337, Neal and Russo state in the introduction that they give an explicit formula for the completely bounded Banach-Mazur distance from H_n^k to $R_n = H_n^n$ and $C_n = H_n^1$, answering a question posed in their earlier paper. Theorem 3 gives

$$d_{cb}(H_n^k, H_n^1) = \sqrt{\frac{kn}{n-k+1}}, \quad 1 \leq k \leq n.$$

Immediately after, a corollary on PDF page 10 says that the following is the answer to Problem 1 in the earlier paper:

$$d_{cb}(H_n^k, H_n^n) = \sqrt{\frac{(n-k+1)n}{k}}, \quad 1 \leq k \leq n.$$

Conclusion. Using $R_n = H_n^n$, the exact answer to the extracted source problem is

$$\boxed{d_{cb}(H_n^k, R_n) = \sqrt{\frac{n(n-k+1)}{k}}} \quad (1 \leq k \leq n).$$

The supporting authors explicitly knew this was the same problem: the later paper says the displayed corollary answers Problem 1 in [1].

Scope. This resolves only the row-distance Problem 1 from the source paper. The broader classification Problem 2 in arXiv:math/0201187 is not settled by this packet, and arXiv:math/0410337 itself poses further distance and duality questions.

References

- [1] M. Neal and B. Russo, *Contractive projections and operator spaces*, arXiv:math/0201187; Trans. Amer. Math. Soc. **355** (2003), 2223–2262.
- [2] M. Neal and B. Russo, *Representation of contractively complemented Hilbertian operator spaces and the Fock space*, arXiv:math/0410337; Proc. Amer. Math. Soc. **134** (2006), 475–485.